

Reviewer Pack — Entropic Complexity of Boolean Functions via Logarithmic Gro...

Entry 12a3e7a6a5ec · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Entropic Complexity of Boolean Functions via Logarithmic Growth Rate

Entry ID: 12a3e7a6a5ec **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-31 04:43:22 UTC

1. Conjecture

Field A (mathematical branch): Entropy Theory in Statistical Mechanics **Field B** (complexity object): Boolean Function Complexity

Statement:

For any given Boolean function f , its entropic complexity $E(f)$ is linearly correlated with the logarithmic growth rate of its number of satisfying assignments, i.e., $E(f) = \Theta(\log(g_n))$, where g_n is the number of satisfying assignments for an instance of size n .

Rationale (proposer's reasoning):

Entropy theory in statistical mechanics has shown that certain physical systems exhibit a correlation between their entropy and their behavior under perturbations. This conjecture suggests that Boolean functions may similarly demonstrate a link between their entropic complexity and the rate at which their satisfiability grows with input size, providing a new perspective on computational complexity.

Taxonomy category: ENTROPY_THEORY (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 589fcfe931349433

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

For the conjecture on entropic complexity of Boolean functions, support is indicated by a correlation coefficient ≥ 0.8 from regression analysis between $E(f)$ and $\log(g_n)$, with no seed producing a metric value > 4 for any statistic.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: ADJACENT_OK against 6 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "entropic complexity" AND "Boolean function complexity" - "logarithmic growth rate" IN BOOLEAN FUNCTIONS AND statistical mechanics" - "number of satisfying assignments" AND entropy theory AND Boolean functions

Top relevant hits considered: - [http://arxiv.org/abs/1902.04564v1] Time's Arrow in a Quantum Universe: On the Status of Statistical Mechanical Probabilities - [http://arxiv.org/abs/1011.3516v1] A statistical-mechanical view on source coding: physical compression and data compression - [http://arxiv.org/abs/2401.02977v2] Open Problems within Nonextensive Statistical Mechanics - [http://arxiv.org/abs/1411.0659v2] Approximate Counting in SMT and Value Estimation for Probabilistic Programs - [http://arxiv.org/abs/2004.14692v1] Sparse Hashing for Scalable Approximate Model Counting: Theory and Practice - [http://arxiv.org/abs/2602.08110v1] Term Coding and Dispersion: A Perfect-vs-Rate Complexity Dichotomy for Information Flow

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math
from fractions import Fraction

def generate_random_boolean_function(n):
    return [random.choice([0, 1]) for _ in range(2**n)]

def count_satisfying_assignments(f, n):
    m = len(f)
    g_n = 0
    for i in range(m):
        if sum(int(x) * int(y) for x, y in zip(bin(i)[2:].zfill(n), f)) == n:
            g_n += 1
    return g_n

def calculate_entropy(g_n):
    total = sum(g_n)
    probabilities = [Fraction(count, total) for count in g_n]
    entropy = -sum(p * math.log2(p) for p in probabilities if p != 0)
    return entropy

def run_trial(seed: int) -> dict:
    random.seed(seed)
    n_values = [5, 10, 15, 20, 30, 40]
    g_n = []
```

```

entropies = []

for n in n_values:
    f = generate_random_boolean_function(n)
    g_n.append(count_satisfying_assignments(f, n))
    entropy = calculate_entropy(g_n)
    entropies.append(entropy)

metric_name = "Entropy"
metric_value = sum(entropies) / len(entropies)
instances_tested = len(n_values)
n_max = max(n_values)
conjecture_holds = all(metric_value <= 4 for _ in range(instances_tested))
counterexample = "" if conjecture_holds else "mapping_undefined"

return {
    "metric_name": metric_name,
    "metric_value": metric_value,
    "instances_tested": instances_tested,
    "n_max": n_max,
    "conjecture_holds": conjecture_holds,
    "counterexample": counterexample
}

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2**i - 1 for i in range(5, 30)]
    results = []

    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_metric_value = sum(r["metric_value"] for r in results) / len(results)
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_metric_value} std=0.0 support_fraction=1.0")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_metric_value} std=0.0 support_fraction={support_fraction}")
    else:
        first_failing_seed = next(i for i, r in enumerate(results) if not r["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample='mapping_undefined' first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_97b7d0fe.py", line 69, in <module>
    result = run_trial(seed)

```

```

^^^^^^^^^^^^^^^^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_97b7d0fe.py", line 44, in run_trial
    entropy = calculate_entropy(g_n)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_97b7d0fe.py", line 31, in calculate_entropy
    probabilities = [Fraction(count, total) for count in g_n]
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "/usr/lib/python3.12/fractions.py", line 281, in __new__
    raise ZeroDivisionError('Fraction(%s, 0)' % numerator)
ZeroDivisionError: Fraction(0, 0)

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test crashed due to a ZeroDivisionError before producing data for regression analysis.
 | next: Investigate the cause of the ZeroDivisionError and ensure that the test can run to completion without errors.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	13516
2	preregistration	ollama_remote	glm4:latest	0	0	9580
3	novelty	ollama_remote	glm4:latest	0	0	8532
4	novelty	ollama_remote	glm4:latest	0	0	19261
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	14365
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	41089
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	19642
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8962
9	judge	ollama_remote	glm4:latest	0	0	14626

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 149573 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/12a3e7a6a5ec.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/12a3e7a6a5ec.jsonl` for verification of provenance

4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: research/pvsnp_sandbox/test_*.py (per-cycle)

Replay tarball: research/replay/12a3e7a6a5ec.tar.gz (if generated)